

SMP Construction

How the kernel, the ports and the boards fit together

SMP construction

How the μ OS++ III SMP kernel, an architecture port, the device drivers and the test applications fit together, and what each one owes the others.

1. The six repositories

micro-os-plus-iii-smp	the kernel, the shared tests, the shared build rules
micro-os-plus-iii-devices	SD, flatfs, DWC2, FatFs, BCM2837 SoC support
micro-os-plus-iii-aarch32	ARMv7-A port
micro-os-plus-iii-aarch64	ARMv8-A port
micro-os-plus-iii-cortexm	ARMv7E-M / ARMv8-M port
micro-os-plus-iii-posix-arch	native host port

Dependencies run **one way**. An architecture project consumes the kernel and, if it needs them, the devices; neither of those knows an architecture project exists. Nothing is vendored: the projects are cloned side by side and CMake finds them as siblings.



Keeping the kernel tree free of device and board code is what lets `src/` and `include/` match upstream micro-os-plus-iii path for path, so the fork can be merged against upstream later with conflicts only in the files that genuinely differ.

2. Division of labour

Concern	Lives in	Why there
Scheduler, threads, sync objects, per-CPU run state	kernel <code>src/rtos/</code>	no machine instructions; identical for every port
Type widths, IRQ state representation, core count	port <code>include/cmsis-plus/rtos/port/os-c-decls.h</code>	the only place a word size is decided
Instructions: barriers, IRQ mask, core id, exclusive monitor	port <code>include/cmsis-plus/rtos/port/os-inlines.h</code>	one file per ISA
Bring-up, kernel lock, context switch hooks	port <code>src/rtos/os-core.cpp</code>	the port's half of the scheduler
Reset vector, MMU, exception vectors, timer	port <code>src/</code>	board and ISA
SD, FatFs, flatfs, USB, SoC mailbox	devices repo	shared verbatim by both ARM ports
Test applications	port <code>test/<board>/</code>	every board owns its own — see <code>tests-in-aarch32-aarch64.md</code>

Two files called `src/rtos/os-core.cpp`

This is deliberate and both are compiled into every image.

- The **kernel's** is the portable SMP scheduler: per-CPU `current_thread_[]`, the affinity picker, the per-core idle bookkeeping. It contains no machine instructions and calls out to `port_cpu_id()`.
- The **port's** is the bring-up and the lock: secondary core release, the recursive kernel spinlock, the context-switch entry points.

They share a path because each is at the path its own project expects; they do not collide, and the predecessor Makefiles compiled both in exactly the same way.

3. The SMP port contract

The whole kernel-side SMP patch is 379 lines and asks a port for a small, closed set of things.

3.1 Types — `cmsis-plus/rtos/port/os-c-decls.h`

```
typedef uint64_t os_port_clock_timestamp_t;
typedef uint32_t os_port_clock_duration_t;
typedef uint64_t os_port_clock_offset_t;
typedef bool     os_port_scheduler_state_t;
typedef uint64_t os_port_irq_state_t;          /* AArch32: uint32_t */
typedef uint64_t os_port_thread_stack_element_t; /* AArch32: uint32_t */
typedef uint64_t os_port_thread_stack_allocation_element_t;

typedef struct { os_port_thread_stack_element_t* stack_ptr; }
                os_port_thread_context_t;

typedef struct { volatile uint32_t lock, owner, depth; } /* kernel lock */
typedef struct { volatile uint32_t lock; }              /* ticket lock */
```

plus the configuration macros in section 5.1.

3.2 Declarations — `cmsis-plus/rtos/port/os-decls.h`

The C++ half: the `port::stack`, `port::interrupts` and `port::scheduler` namespaces, and the four objects a port must **define**:

```
extern volatile state_t  lock_state[OS_NCPU];
extern smp_klock_t       _smp_klock;
extern smp_tlock_t       _smp_tlock;
extern volatile unsigned _port_ctx_pending[OS_NCPU];
```

This file contains no machine instructions, and the two AArch ports had it byte-identical but for one missing `volatile` (cortexm keeps its own, larger `os-decls.h`). It therefore lives **once**, in the kernel repository at `port/smp-common/`, and a port opts in:

```
target_link_libraries (my-port INTERFACE micro-os-plus::port-smp-decls)
```

It is on no default include path. A port with different needs writes its own `os-decls.h` in its own `include/` and simply does not link that target, so the two can never be confused.

The kernel's `include/cmsis-plus/rtos/os-decls.h` also declares the port's `port::clock_highres` contract (`has_hardware_counter()` and `hardware_counter()`), while `include/cmsis-plus/rtos/os-thread.h` defines `thread::state::destroying = 7` to arbitrate SMP thread teardown between `thread::kill()` and `os_rtos_idle_actions()`.

3.3 Inline operations — `cmsis-plus/rtos/port/os-inlines.h`

Per ISA. The kernel calls, and the port supplies:

Function	AArch32	AArch64
<code>port_cpu_id_inline()</code>	<code>mrc p15, 0, Rd, c0, c0, 5</code>	<code>mrs Xd, mpidr_el1</code>
<code>interrupts::critical_section::enter/exit</code>	<code>cpsid i</code> / <code>cpsie i</code>	<code>msr daifset/daifclr, #2</code>
<code>interrupts::uncritical_section::enter/exit</code>	as above, inverted	as above, inverted
<code>scheduler::lock/unlock/locked</code>	recursive kernel lock	recursive kernel lock
<code>_smp_klock_raw_acquire/release</code>	<code>ldrex</code> / <code>strex</code>	<code>ldaxr</code> / <code>stlxr</code>
<code>clock_highres::has_hardware_counter()</code>	<code>true</code>	<code>true</code>
<code>clock_highres::hardware_counter()</code>	<code>timer_arm::get_count()</code> (<code>CNTPCT</code>)	<code>timer_arm::get_count()</code> (<code>CNTPCT_EL0</code>)

3.4 Functions with C linkage

```

unsigned port_cpu_id (void);          /* current core index, 0..OS_NCPU-1 */
void      port_sys_init (void);        /* per-core tick + IRQ controller */
void      port_ctx_switch (void);      /* context-switch entry */
void      port_smp_ipi (unsigned cpu); /* send reschedule IPI to target CPU */
void      port_smp_secondary_start (void); /* secondary core entry point */
void      smp_install_boot_threads (void); /* supplied by the application */

```

`smp_install_boot_threads()` is the one the *application* owes the port: it creates an idle thread per secondary CPU and registers each in `os::rtos::scheduler::os_idle_thread_core[]` before the cores are released. Most of a board's tests get it from that board's `test/<board>/src/test-smp-boot.cpp`; a test carried over whole from the predecessor brings its own, and says so through `BOARD_TEST_SELF_CONTAINED` in the board's `test/<board>/tests.cmake`.

3.5 Bring-up order

```

reset -> startup.S          core 0 only; others park
        -> os_main          application
        -> interrupts::uncritical_section::enter ()  unmask IRQs
        -> smp_install_boot_threads ()              idle thread per core
        -> smp::start_secondary_cores ()            release cores 1..N-1
        -> scheduler::start ()

```

Each secondary enters `port_smp_secondary_start()`, takes the kernel lock, initialises its tick through `port_sys_init()`, and schedules its idle thread.

4. What makes the test applications ISA-neutral

One copy of each test is compiled by every port that can run it. The disagreements between the old 32b and 64b copies resolved like this:

Was	Now	Note
<code>dsb</code> / <code>dmb</code> vs <code>dsb sy</code> / <code>dmb ish</code>	<code>dsb sy</code> , <code>dmb ish</code>	not a difference: the explicit forms assemble on ARMv7-A as well
<code>cpsie i</code> vs <code>msr daifclr, #2</code>	<code>interrupts::uncritical_section::enter()</code>	a kernel API every port implements
<code>mrrc p15,...,c14</code> vs <code>mrs cntpct_el0</code>	<code>timer_arm::get_count()</code>	the port already had it
"(AArch64)" in a banner	<code>PORT_BANNER_ISA</code>	supplied by the port's <code>uart.hpp</code>

The rule: **an application names no register and no instruction that is not portable across the ports that can run it.** What it needs from a port it takes through the port's headers.

5. Preprocessor defines, per folder

5.1 `micro-os-plus-iii-aarch32` / `-aarch64` — defined by the port header

Set in `include/cmsis-plus/rtos/port/os-c-decls.h`; the build may override the first two.

Macro	Value	Meaning
<code>OS_NCPU</code>	4 (<code>#ifndef</code>)	core count; the build chooses
<code>OS_USE_SMP_SCHEDULER</code>	1 when <code>OS_NCPU > 1</code>	selects the SMP scheduler over the scalar one
<code>SMP_NO_OWNER</code>	<code>0xFFFFFFFFu</code>	"kernel lock unowned" sentinel
<code>OS_SMP_IPI_SGI</code>	0	cross-core reschedule signal; BCM2837 has no GIC, so local mailbox 0
<code>OS_INTEGER_RTOS_STACK_FILL_MAGIC</code>	<code>0xEFBEADDE</code> / <code>...EFBEADDEULL</code>	stack paint; element width
<code>OS_HAS_INTERRUPTS_STACK</code>	defined	the port provides a separate IRQ stack

5.2 Set by the build for every application

`uos_add_app()` in `cmake/uos-app.cmake`:

Macro	Source
<code>OS_NCPU=<n></code>	<code>NCPU</code> argument
<code>OS_USE_SMP_SCHEDULER=1</code>	added automatically when <code>NCPU > 1</code>

The architecture project's `test/CMakeLists.txt`:

Macro	Where	Meaning
<code>TRACE</code>	both	enable the trace channel
<code>__ARM_EABI__</code>	both	ARM EABI
<code>__ARM_ARCH_7A__</code>	aarch32 only	ARMv7-A
<code>SEMIHOST</code>	both	mirror the console to the semihosting channel and exit through it once a RESULT is printed. On by default , as it was in the Makefiles, so no build is silently non-semihosting
<code>BOARD_RPI3B</code>	<code>-DBOARD=rpi3b</code>	Pi 3 B instead of Pi Zero 2 W: other linker script, other banner
<code>HW_BUILD</code>	<code>hwd</code> variant	SD tests use the existing FAT32 boot partition through FatFs instead of formatting a blank card
<code>DEBUG_BOOT</code>	<code>-DUOS_DEBUG_BOOT=ON</code>	early-boot asm markers from <code>startup.S</code> , for bring-up under OpenOCD. Off by default , as the sources assume: each marker is a semihosting trap, and under a JTAG probe a trap costs real time
<code>LED_PIN</code>	<code>BOARD=rpi-zero-2w</code>	the GPIO driven as the user LED, 29 by default — the Zero 2 W's onboard green ACT LED. <code>led.hpp</code> 's own fallback is GPIO 16, header pin 36, which blinks nothing on a bare board. Which pin it is depends on the board , so it is set here for every test rather than per application; <code>BOARD=rpi3b</code> does not use it at all, because there the ACT LED is VideoCore expander GPIO 130 behind the mailbox

5.3 Per-application, from the board's `test/<board>/tests.cmake`

On the Pi boards two applications have knobs. `cmsis-os-validator` gets `UOS_CMSIS_OS_VALIDATOR`, which makes the board's sources map a "no cycle counter" page where the validator probes the Cortex-M DWT and route the local Mailbox 1 interrupt to the validator's handler (its stand-in for NVIC IRQ 0); no other image changes. `usb_test` has these:

Macro	Default	Meaning
<code>LED_BLINKS</code>	3	blinks per burst
<code>LED_ON_MS</code> / <code>LED_OFF_MS</code>	40 / 40	half-cycles; the tick is 1 ms, so each is clamped to at least that
<code>LED_GAP_MS</code>	300	dark gap between bursts
<code>USB_FORCE_FS</code>	defined	force full speed. It also sets the driver's <code>kMpsHs</code> to 64, which is what bounds a bulk transfer: <code>D{I,0}EPTSIZ.PKTCNT</code> is 10 bits on this core, so 1023 packets — 65472 bytes at full speed against 523776 at high speed

Consumed but not set by default:

Macro	Meaning
<code>RUN_MS</code>	wall-clock run length for <code>smp-num-test</code> , <code>smp-pipeline-test</code> , <code>smp-pro-cons-test</code> . The sources carry their own defaults; set it for short QEMU smoke runs
<code>TEST_IDLE_STACK_WORDS</code>	stack words per secondary idle thread, default 512

5.4 `micro-os-plus-iii-smp` — the kernel

The kernel reads a large set of optional `OS_*` switches (`OS_TRACE_RTOS_*`, `OS_INCLUDE_RTOS_STATISTICS_*`, `OS_USE_RTOS_PORT_*`, ...). **None is required.** The ARM builds set none of them; every one defaults to off.

Two are structural:

Macro	Meaning
<code>OS_USE_OS_APP_CONFIG_H</code>	every kernel TU includes <code><cmsis-plus/os-app-config.h></code> from the port. The ARM ports keep that file empty and do not define this
<code>OS_USE_SMP_SCHEDULER</code>	selects the SMP scheduler. Comes from the port header or the build, never hand-written in a source

5.5 `micro-os-plus-iii-devices`

FatFs is configured entirely through `fatfs/ffconf.h` (`FF_*`, ~40 macros); nothing needs to be passed on the command line. The only build-visible macro is `USB_FORCE_FS`, which `usb_test` sets.

Device sources dispatch on `__aarch64__` inline where an instruction genuinely differs — three places in total, for the cache maintenance and the core-id read. No project-specific macro selects an ISA.

6. Verifying a port

```
tools/verify-kernel-compiles.sh [--without <group>]... \  
  <port-include-dir>[:<dir>...] [compiler [flags...]]
```

Several directories, `:`-separated, are searched in that order: a port whose board overlays one on another (cortexm's `include-rp2350:include`), or the AArch ports with the kernel's `port/smp-common`. A bare-metal port passes its cross compiler, followed by the facts a board would supply (aarch32: `-mcpu`, `OS_NCPU`, `OS_SMP_IPI_SGI`, `PORT_GREETING`). `--without startup` skips the optional `startup` group, which the AArch ports do not link; the skipped sources are reported, never counted. `.c` sources compile as C.

Compiles every source the kernel declares against that port's headers. It is the only meaningful standalone check: the kernel can never compile on its own, because `include/cmsis-plus/rtos/os-decls.h` includes `<cmsis-plus/rtos/port/os-decls.h>`, which only a port supplies.

Current result on `posix-arch` and `aarch64`: **61 declared, 61 compiled**; on `aarch32`, 57 compiled and the 4 `startup` sources skipped. The exact commands are in `STATUS.md`.